

100 Days of Linux - Day 036

Title: Transforming Text with tr

Today's Goal

Learn how to translate, replace, delete, and squeeze characters using the **tr** command.

Why This Matters

Data engineers often clean raw data before processing it. The **tr** command is useful for changing text formats, removing unwanted characters, and standardizing data.

Command of the Day: tr

Syntax: `cat file | tr 'SET1' 'SET2'`

Common Examples:

`cat names.txt | tr 'a-z' 'A-Z'` (lowercase to uppercase)

`cat names.txt | tr 'A-Z' 'a-z'` (uppercase to lowercase)

`cat data.txt | tr -d ','` (delete commas)

`cat text.txt | tr -s ' '` (squeeze repeated spaces)

Explanation: The **tr** command reads text from standard input and transforms characters.

Beginner Lesson

Unlike many commands, **tr** does not read a filename directly. It works with input from another command, usually through a pipe (`|`). This introduces an important Linux concept: connecting commands together.

Practice Questions

1. Create a file named `names.txt` with lowercase names. Convert them to uppercase using **tr**.
2. Create a file with uppercase text and convert it to lowercase.
3. Create a file containing commas and remove all commas using `tr -d`.
4. Create a file with multiple consecutive spaces. Use `tr -s` to squeeze repeated spaces into a single space.
5. Display your current working directory after completing today's tasks.

Hints

Remember that **tr works with input from another command, such as `cat filename | tr ...`**

Mini Challenge

Create `employees.csv` with inconsistent capitalization and extra commas or spaces. Use **tr to convert names to uppercase, remove commas from a copy, and squeeze repeated spaces.**

Common Mistakes

Trying to run `tr` directly on a filename. Use a pipe (`|`) to send file contents into `tr`.

Did You Know?

The `tr` command only transforms characters. It does not edit the original file unless you redirect the output to a new file.

Pro Tip

Combine `tr` with `grep`, `cut`, and `sort` to build powerful text-cleaning pipelines.

Answer Key (Instructor Only)

Q1: `cat names.txt | tr 'a-z' 'A-Z'`

Q2: `cat names.txt | tr 'A-Z' 'a-z'`

Q3: `cat data.txt | tr -d ','`

Q4: `cat text.txt | tr -s ' '`

Q5: `pwd`

Homework

Practice every `tr` option on three different text files and save the transformed output to new files using `>`.

Next Day Preview

Tomorrow you'll learn the `sed` command to search, replace, and edit text automatically.